



Prob. 1: For the questions a, b, c, d & e consider the circuit shown in figure 1 and answer (use nodal analysis method):

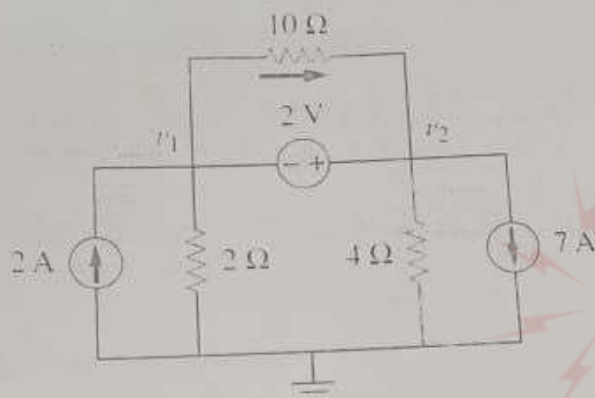


Fig.1

a. The value of V_1 is:

-7.333 [V]

2 marks

b. The value of V_2 is:

-5.333 [V]

2 marks

c. The value of the current (i) through 10Ω resistor is:

-0.2 A

1 mark

d. The power absorbed by 4Ω resistor is:

7.11 Watts

1 mark

e. The power absorbed by 2 A current source is:

14.666 Watts

1 mark

f. Write the required equations (do not simplify)

(2V source + 10Ω) form supernode: $2 = \frac{V_1}{2} + \frac{V_2}{4} + 7$
 $-2 = V_1 - 2V_2$

2 marks

1 mark

Prob. 2: Consider the circuit shown in figure 2 and answer:

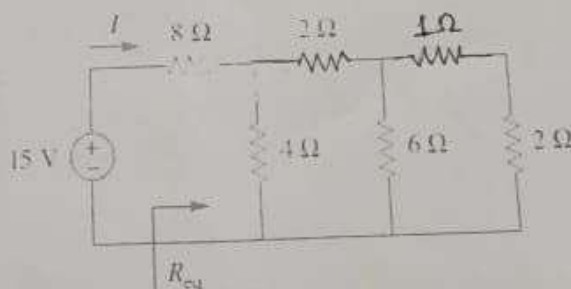


Fig.2

a. The equivalent resistance R_{eq} is:

$R_{eq} = 10\Omega$

3 marks



a. The current I is:

$$I = \frac{15}{10} = 1.5 \text{ A}$$

3 marks

The Power absorbed by 4Ω resistor is:

$$(0.75)^2 \times 4 = 2.25 \text{ Watt}$$

2 marks

b.3: a. The charge transferred through an element is given by, $q(t) = 2t^2 + 5t^2 + 6$ C. Calculate the current entering the element at instant $t = 10$ ms.

3 marks

$$i = \frac{dq}{dt} = \frac{d(2t^2 + 5t^2 + 6)}{dt} = 10t \text{ A}$$

$$i = 10 \times 10^{-3} = 100 \times 10^{-3} = 100 \text{ mA}$$

A load draws 500W when connected to a 230 V line. Calculate the current and the energy consumed in 2 hours in joules?

$$1) I = \frac{500}{230} = 2.174 \text{ A} ; 2) W = P \cdot t = 500 \times 2 \times 60 \times 60 \text{ J}$$

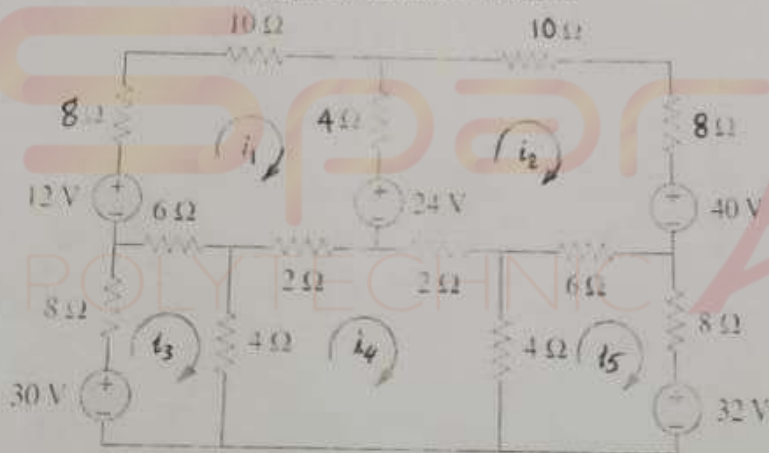
$$= 1000 \times 3600 \text{ J}$$

$$= 3600 \text{ kJ}$$

$$= 3.6 \text{ MJ}$$

3 marks

c. Refer to figure 3; write the mesh equations in matrix form:



$$[A][i] = [V] \quad \text{--- Fig. 3}$$

2 marks

$$\begin{bmatrix} 30 & -4 & -6 & -2 & 0 \\ -4 & 30 & 0 & -2 & -6 \\ -6 & 0 & 18 & -4 & 0 \\ -2 & -2 & -4 & 12 & -4 \\ 0 & -6 & 0 & -4 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} -12 \\ -16 \\ 30 \\ 0 \\ -32 \end{bmatrix}$$

4 marks